

DR-2 lattice closure mapped onto STEP-5B G1-triple-prime-AE – the

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1. Purpose and scope

R-026 closes the (unweighted) additive-energy DR-2 for the crystallographic lattice class at T7. STEP-5B is CLOSED-CONDITIONAL on H-ADM-COH, whose role is to make the G1'''-AE corner – the weighted sphere additive-energy bound – hold by restricting to angularly SEPARATED competitor patterns. This note maps the lattice closure onto G1'''-AE through an amplitude bridge and analyses what remains for an H-ADM-COH discharge. It introduces no new physics; it is a reduction/integration note.

2. Three identifications

(i) **G1'''-AE is a weighted additive energy.** The STEP-5B corner requires $\sum_t w_t^2 \leq K(\lambda' I)^2$ with $w_t = \sum_{a+b=t} c_a c_b$ the amplitude-weighted representation function of the pattern points a, b on the shell and $I = \|c\|_2^2 = \sum_a |c_a|^2$. This is the weighted additive energy of the pattern.

(ii) **The “carrier” is the sum-level circle.** A pair (a, b) with $a + b = t$ is diametral on its carrier circle of centre $t/2$; that carrier is exactly the sum-level circle $C_t = \{x : |x|^2 = R, x \cdot t = \frac{1}{2}|t|^2\}$ of R-025. The carrier occupancy is $\#(Q \cap C_t)$, whose maximum is $T'(Q)$, and the extraction-route carrier-richness $\chi(P)$ is the dual incidence quantity (rich carriers through a point).

(iii) **H-ADM-COH is angular separation.** H-ADM-COH admits competitor patterns whose pairwise angular separations are $\geq \theta_{\min}(\hat{r})$; finer-scale mass is assigned to the Gaussian sea. This is the SEPARATED-pattern restriction – the regime in which the additive-energy bound is classically easy. Removing H-ADM-COH is the unrestricted (non-separated) problem.

3. The amplitude bridge (WEIGHTED Lemma A)

Lemma (weighted). For finite $Q \subset S^2$ and any amplitudes $c = (c_a)_{a \in Q}$,

$$\boxed{\sum_t |w_t|^2 \leq (1 + T'(Q)) I^2}, \quad w_t = \sum_{a+b=t} c_a c_b, \quad I = \sum_a |c_a|^2.$$

Proof. Split $\sum_t |w_t|^2 = |w_0|^2 + \sum_{t \neq 0} |w_t|^2$. For $t = 0$, $w_0 = \sum_{a: -a \in Q} c_a c_{-a}$, so by Cauchy–Schwarz $|w_0| \leq \sum_a |c_a| |c_{-a}| \leq (\sum_a |c_a|^2)^{1/2} (\sum_a |c_{-a}|^2)^{1/2} = I$, i.e. $|w_0|^2 \leq I^2$. For $t \neq 0$, Cauchy–Schwarz over the $r(t)$ summands gives $|w_t|^2 \leq r(t) \sum_{a+b=t} |c_a|^2 |c_b|^2$, and $r(t) \leq \#(Q \cap C_t) \leq T'(Q)$, so

$$\sum_{t \neq 0} |w_t|^2 \leq T'(Q) \sum_{t \neq 0} \sum_{a+b=t} |c_a|^2 |c_b|^2 \leq T'(Q) \sum_{a,b} |c_a|^2 |c_b|^2 = T'(Q) I^2.$$

Adding the two parts gives the bound. $\square$

This is the amplitude-weighted analogue of Lemma A (R-025): the SAME sum-circle richness T' controls the weighted energy, for arbitrary amplitudes – so the amplitude degree of freedom of G1''-AE does not require new geometry.

4. The integration bound

Theorem. Let $Q \subseteq \Lambda \cap \{|x|^2 = R\}$ be any subset of a crystallographic momentum shell, and c any amplitudes. Then

$$\sum_t w_t^2 \leq (1 + 6 \max_{t \neq 0} d(4R - |t|^2)) \|c\|_2^4 \leq (1 + C_\epsilon R^\epsilon) \|c\|_2^4.$$

Equivalently, $\sum_t |w_t|^2 \leq K I^2$ with $K = 1 + C_\epsilon R^\epsilon$ – the G1''-AE bound with a SUBPOLYNOMIAL constant, holding for ALL subsets of the shell (no angular separation).

Proof. WEIGHTED Lemma A bounds $\sum_t |w_t|^2$ by $(1 + T'(Q))I^2$; R-026 ([DIV-CIRC]) bounds $T'(Q) \leq 6 \max_{t \neq 0} d(4R - |t|^2) \ll_\epsilon R^\epsilon$. $\square$

5. What this means for H-ADM-COH (operator decision)

The bound of §4 is exactly the G1''-AE requirement, established for the UNRESTRICTED (non-separated) lattice class. Since H-ADM-COH was the angular-separation restriction that secured G1''-AE, the lattice arithmetic REPLACES separation: for the physical (crystallographic-shell) competitor class, the weighted additive-energy bound holds without H-ADM-COH. The discharge is nonetheless NOT enacted here; three residuals, all operator-level, remain:

- (a) *Subpolynomial vs constant K.* §4 gives $K = 1 + C_\epsilon R^\epsilon$, not a fixed constant. With $R \asymp N^2$, $K \sim N^{2\epsilon}$ (measured $K = 1 + T'$ grows like $R^{0.126}$). This matches the pre-registered sharp- $O(N^2)$ conjecture (exponents 2.04–2.08) and is far inside the $\times 55.6 / \times 8.8 / \times 2.1$ margins for the operating n , but whether the gate accepts a subpolynomial K in place of the constant $K = 14$ form is the gate-keeper's call.
- (b) *Modeling identification.* The discharge requires that admissible competitor patterns are exactly subsets of the crystallographic momentum shell (physical for a lattice field theory); this is an operator modeling confirmation, not a mathematical step.
- (c) *Corner coverage.* The G1''-AE corner (non-transversal multi-circle high- n) is a lattice-shell subset, hence covered by §4; the operator confirms no corner escapes the lattice identification.

If (a)–(c) are accepted, H-ADM-COH may be discharged for the lattice class and B1's hypothesis set reduced accordingly. This note makes NO such flip: DR2-SHARE stays OPEN, B5 T5, B1 T6, H-ADM-COH retained.

6. Devil's-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

- (α) “WEIGHTED Lemma A might fail for signed/complex amplitudes.” Checked: the Cauchy–Schwarz proof is amplitude-agnostic; verified on uniform, Gaussian, peaked, and signed profiles (worst ratio $0.277 \leq 1$, dr2_weighted_energy.py 3/3).

- (β) “*This silently flips H-ADM-COH.*” It does NOT. §5 lists three operator-level residuals and makes no gate/tier/hypothesis change. The note’s theorems (WEIGHTED Lemma A, the §4 bound) are T7; the discharge is deferred.
- (γ) “*K is subpolynomial, so this is weaker than the constant-K G1’’-AE.*” VALID and recorded as residual (a). A subpolynomial K is consistent with the sharp- $O(N^2)$ conjecture and the measured exponents; whether it suffices for the margin closure is the operator’s call, not a mathematical defect of the bound.
- (δ) “*The $\chi(P)$ route is bypassed – is that legitimate?*” YES: the extraction route’s $\chi(P)$ obstruction (DR2-SHARE) was internal to one proof strategy. The lattice bound proves the additive energy DIRECTLY (Lemma A + divisor bound), so the weighted energy is controlled without ever iterating the extraction; $\chi(P)$ is not needed. DR2-SHARE (the extraction obstruction) is thereby rendered moot FOR THE LATTICE CLASS, though it stays formally OPEN.

Quantitative sanity check (mandatory). WEIGHTED Lemma A verified exactly: $\sum_t w_t^2 \leq (1 + T') \|c\|_2^4$ on shells $R = 101, 314, 561$, four amplitude profiles, worst ratio 0.277. Antipodal term $w_0^2 \leq \|c\|_2^4$ checked. Subpolynomial K : $1 + T'$ grows like $R^{0.126} < R^{0.30}$. Direction: upper bound, $\|c\|_2^2 = I > 0$.

7. Result footer

Result ID:	R-027 / DR-2 -> STEP-5B G1’’-AE integration
Precise statement:	WEIGHTED Lemma A: for finite Q in S^2 and any amplitudes c , $\sum_t w_t^2 \leq (1+T'(Q)) \ c\ _2^4$, $w_t = \sum_{\{a+b=t\}} c_a c_b$ (Cauchy-Schwarz; $t=0$ split). With R-026 (lattice $T' \ll_{\text{eps}} R^{\text{eps}}$): for any subset Q of a crystallographic shell, $\sum_t w_t^2 \leq (1+C_{\text{eps}} R^{\text{eps}}) \ c\ _2^4$ -- the G1’’-AE bound with $K \sim R^{\text{eps}}$, NO angular separation.
Scope:	STEP-5B G1’’-AE corner; the DR2-SHARE integration; the physical (lattice) competitor class.
Dependencies:	WEIGHTED Lemma A (T7, this note) + R-026 (T7). Both unconditional modulo textbook number theory.
Evidence grade:	T7 (weighted Lemma A + integration bound). The H-ADM-COH discharge is NOT enacted (operator decision, residuals (a)-(c)).
Reproduction command:	python codes/vacuum/dr2_weighted_energy.py
Expected output:	3/3 PASS; $\sum_t w_t^2 \leq (1+T') \ c\ _2^4$ (worst ratio 0.277); $w_0^2 \leq \ c\ _2^4$; $K=1+T' \sim R^{0.126}$.
Falsification gate:	Amplitudes c with $\sum_t w_t^2 > (1+T'(Q)) \ c\ _2^4$ (refutes WEIGHTED Lemma A); not observed.
Tier before / after:	NEW T7 integration result. NO flip: H-ADM-COH retained, DR2-SHARE OPEN, B5 T5, B1 T6.
No-overclaim statement:	Establishes the weighted G1’’-AE bound for the lattice class with subpolynomial K , decoupling-free. Does NOT discharge H-ADM-COH or change any tier; residuals (a) subpolynomial- K acceptance, (b) modeling identification, (c) corner coverage are operator decisions. Arbitrary- Q (non-lattice) DR-2 stays OPEN.
Next required action:	Operator: accept/deny residuals (a)-(c); if accepted, discharge H-ADM-COH for the lattice class and re-tier B1.